

ATHUL DEV

Backend Systems Engineer | Scalable APIs & Async Architecture

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SUMMARY

Backend Systems Engineer specializing in asynchronous API architecture, high-concurrency database design, and performance profiling. Builds production FastAPI microservices across PostgreSQL and MongoDB, resolving performance bottlenecks, memory/connection pooling leaks, and security risks under load. Reduced API latency 66% via load testing and query optimization, and engineered multi-layer HMAC/OAuth authentication and idempotent webhook workflows.

SKILLS

- **Backend Architecture:** FastAPI, RESTful APIs (Application Programming Interfaces), WebSockets, Microservices, Event-Driven Architecture, Connection Pooling, Asynchronous Workflows
- **Programming & Core:** Python (Asyncio), SQL, Data Structures, Linux, Git
- **Databases & Storage:** PostgreSQL (Neon, Aiven), MongoDB (Motor, Beanie), SQLite, Redis, asyncpg connection pooling
- **Security & Reliability:** HMAC signature verification, JWT (JSON Web Tokens) / OAuth 2.0, idempotent handling, load testing (Locust), query profiling (EXPLAIN ANALYZE)
- **Cloud & DevOps:** FastAPI Cloud, Docker, APScheduler, QStash, Webhooks, Cloudflare Tunnels, Railway, Cloudinary CDN
- **AI Systems Integration:** LLM (Large Language Model) Agents, Tool Calling, RAG (Retrieval-Augmented Generation), ChromaDB, Multi-Provider Fallback Architecture

PROFESSIONAL EXPERIENCE

Web Developer & AI Engineer — OquiliX

Kochi, India | July 2025 – Present

- Architected production-grade FastAPI REST APIs for e-commerce inventory and order processing across concurrent user sessions.
- Designed asynchronous database architectures supporting non-blocking read/write operations and improved reliability under high-traffic loads.
- Improved AI query relevance through vector-based semantic search pipelines, reducing manual data-retrieval overhead.

ENGINEERING SYSTEMS PROJECTS

Neolix — B2B Sales Outreach Automation

Feb 2026 | neolix-sage.vercel.app

- Reduced median API latency 66% (2700ms → 920ms) by diagnosing a per-request database connection bottleneck via Locust load testing and implementing asyncpg connection pooling; validated across four independent runs with sub-millisecond GIN-indexed full-text search over a 1M+ profile dataset.
- Fixed a cross-database architecture bug where a PostgreSQL query referenced a non-existent dedup table, replacing it with a MongoDB-backed lookup consistent with system design.
- Built a 9-day, 3-channel (Email/WhatsApp/SMS) outreach cadence engine with cost-latency AI copywriting (Groq Llama 3.3 70B/8B), Gmail API integration, and a native Android SMS gateway syncing local logs.

Ekabhumi — E-Commerce Platform

July 2025 | www.ekabhumi.in

- Designed a three-layer payment verification system: client-side HMAC-SHA256 signature check, independent server-side Razorpay API cross-check with Decimal recomputation, and separately-keyed webhook verification; used idempotent processing to block duplicate fulfillment.
- Built Google OAuth 2.0 authentication issuing self-signed, issuer-scoped JWTs with server-side role assignment and silent-refresh flow, avoiding reliance on stored third-party tokens or session cookies.

DOOM — Personal AI Assistant

Nov 2025 | doom-1a9d9743.fastapicloud.dev

- Architected a ReAct-style agent orchestrating **35 specialized tools** behind strict tool-calling protocols and hallucination guardrails preventing false claims.
- Achieved zero-downtime AI availability through a **4-provider, 14-model fallback chain** (Gemini → OpenAI → Groq → OpenRouter), handling provider rate limits and outage events.
- Built a ChromaDB RAG pipeline with overlap-aware chunking and re-ingestion deduplication automating workflows across Naukri, LinkedIn, Snov.io, and Gmail API.

EDUCATION

Bachelor of Computer Applications (BCA) — University of Calicut

2022 – 2025